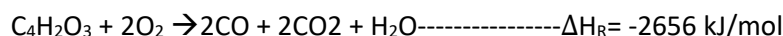
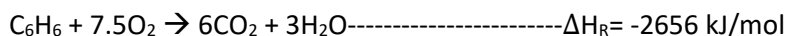
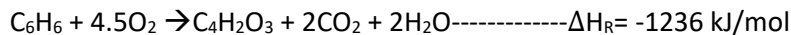


Matt Egeler 321 Project

Goal: Design a reactor to produce Maleic-anhydride($C_4H_2O_3$) from benzene(C_6H_6)

Chemical Equation:



Reaction Rates:

$$r_1 = \frac{k_1 P_1^{\alpha_1}}{(1 + K_2 P_2^2)}$$

$$r_2 = k_2 P_1^{\alpha_2}$$

$$r_3 = \frac{k_3 P_2}{(1 + K_2 P_2^2)}$$

Where,

$$k_1 = 1 \times 10^{-6} \text{ kmol/kg} \cdot \text{s} \cdot \text{atm}^{0.35}$$

$$k_2 = 1.5 \times 10^{-7} \text{ kmol/kg} \cdot \text{s} \cdot \text{atm}^{0.35}$$

$$k_3 = 2.9 \times 10^{-6} \text{ kmol/kg} \cdot \text{s} \cdot \text{atm}$$

$$\alpha_1 = \alpha_2 = 0.35$$

$$K_2 = 310 \text{ atm}^{-1}$$

Converting to SI units gives,

$$k_1 = \frac{1 \times 10^{-6} \text{ kmol}}{\text{kg} \cdot \text{s} \cdot \text{atm}^{0.35}} \cdot \frac{1 \text{ atm}^{0.35}}{101325^{0.35} \left(\frac{\text{N}}{\text{m}^2}\right)^{0.35}} \cdot \frac{1000 \text{ mol}}{1 \text{ kmol}} = \boxed{\frac{1.77 \times 10^{-5} \text{ mol}}{\text{kg} \cdot \text{s} \cdot \left(\frac{\text{N}}{\text{m}^2}\right)^{0.35}}}$$

$$k_2 = \frac{1.5 \times 10^{-7} \text{ kmol}}{\text{kg} \cdot \text{s} \cdot \text{atm}^{0.35}} \cdot \frac{1 \text{ atm}^{0.35}}{101325^{0.35} \left(\frac{\text{N}}{\text{m}^2}\right)^{0.35}} \cdot \frac{1000 \text{ mol}}{1 \text{ kmol}} = \boxed{\frac{2.66 \times 10^{-6} \text{ mol}}{\text{kg} \cdot \text{s} \cdot \left(\frac{\text{N}}{\text{m}^2}\right)^{0.35}}}$$

$$k_3 = \frac{2.9 \times 10^{-6} \text{ kmol}}{\text{kg} \cdot \text{s} \cdot \text{atm}} \cdot \frac{1 \text{ atm}}{101325 \frac{\text{N}}{\text{m}^2}} \cdot \frac{1000 \text{ mol}}{1 \text{ kmol}} = \boxed{\frac{2.86 \times 10^{-8} \text{ mol}}{\text{kg} \cdot \text{s} \cdot \text{atm}}}$$

To find k at different equations we use,

$$k(T) = k(T_0) \exp(E_1/R \cdot (1/T_1 - 1/T))$$

Giving,

$$k_1 = 1.77 \times 10^{-5} \cdot e^{\frac{83100}{8.314} \cdot \left(\frac{1}{673.15} - \frac{1}{T}\right)}$$

$$k_2 = 2.66 * 10^{-6} * e^{\frac{83100}{8.314} * (\frac{1}{673.15} - \frac{1}{T})}$$

$$k_3 = 2.86 * 10^{-8} * e^{\frac{155000}{8.314} * (\frac{1}{673.15} - \frac{1}{T})}$$

Mole Balances:

$$\frac{dFC_6H_6}{dW} = r'_1 + r'_2$$

$$\frac{dFO_2}{dW} = 4.5r'_1 + 7.5r'_2 + 2r'_3$$

$$\frac{dFC_3H_2O_3}{dW} = -r'_1 + r'_3$$

$$\frac{dFH_2O}{dW} = -2r'_1 - 3r'_2 - r'_3$$

$$\frac{dFCO_2}{dW} = -2r'_1 - 6r'_2 - 2r'_3$$

$$\frac{dFCO}{dW} = -2r'_3$$

$$\frac{dFN_2}{dW} = 0$$

$$F_T = F_{C_6H_6} + F_{O_2} + F_{C_3H_2O_3} + F_{H_2O} + F_{CO_2} + F_{CO} + F_{N_2}$$

To find the pressure drop, we will need to use Dalton's law of partial pressure:

$$P_1 = \frac{F_{C_6H_6}}{F_{T_0}} P \text{-----For Benzene}$$

$$P_2 = \frac{F_{C_3H_2O_3}}{F_{T_0}} P \text{-----For Maleic-anhydride}$$

$$F_{T_0} = \frac{.1 \text{ kmol}}{\text{hr}} \left(\frac{1000 \text{ mol}}{1 \text{ kmol}} \right) \left(\frac{1 \text{ hr}}{3600 \text{ s}} \right) = \frac{.02778 \text{ mol}}{\text{s}}$$

$$F_{C_6H_6} = (.15 * .02778) = \frac{0.004167 \text{ mol}}{\text{s}}$$

$$F_{O_2} = F_{N_2} = 0.02778 * \frac{(1 - .15)}{2} = \frac{0.01181 \text{ mol}}{\text{s}}$$

To find the pressure drop, we use the Ergun equation:

$$\frac{dP}{dW} = - \frac{\frac{G(1-\phi)}{\rho_o * g_c * D_p * \phi^3} \left[150 * \frac{(1-\phi)\mu}{D_p} + 1.75G \right]}{\rho_c A_c} * \frac{P_o}{P} \left(\frac{T}{T_o} \right) \left(\frac{F_T}{F_{T_0}} \right)$$

where,

$$B_o = \frac{G(1-\phi)}{\rho_o * g_c * D_p * \phi^3} \left[150 * \frac{(1-\phi)\mu}{D_p} + 1.75G \right]$$

$$\rho_o = \frac{(3.2 * 10^3)kg}{m^3}, g_c = 1, \varphi = .45, D_p = .002m, G = \rho u, u = \frac{v_o}{A_c}, \rho = \frac{\rho_o P}{P_o} \left(\frac{T_o}{T} \right) * \left(\frac{F_{To}}{F_T} \right)$$

From ideal gas law

$$v_o = \left(\frac{F_{To} * R * T_o}{P_o} \right) = \frac{.02778 * 8.314 * 573.15}{2atm} \left(\frac{1atm}{\frac{101325N}{m^2}} \right) = 6.532 * 10^{-4} \left(\frac{m^3}{s} \right)$$

to find A_c we need the inner diameter of the pipe,

$$D_{inner} = 25 * 10^{-3}m - \left(2 * \frac{1}{8}in * \frac{2.54cm}{1in} * \frac{1m}{100cm} \right) = .01865m$$

$$u = \frac{6.532 * 10^{-4}}{\frac{\pi}{4} * (.01865)^2} = \frac{2.39m}{s}$$

Energy balance for a PBR with multiple reactions:

$$\frac{dT}{dW} = \frac{Ua(T_a - T) + \sum_1^3 r'_i * [-\Delta H_{Rxn}(T)]}{\rho_b \sum_1^7 F_j C_{pj}}$$

$$\frac{Ua}{\rho_b} = \frac{120W}{m^2K} * \left(\frac{4}{25 * 10^{-3}m} \right) \left(\frac{1m^3}{3.2 * 10^3kg} \right) = \frac{6W}{K}$$

C_{pi} (J/molK) data from NIST website using Shomate Equation

$$C_{pC_6H_6}(300^\circ C) = 83.02$$

$$C_{pO_2} = 31.32234 - 20.23531T + 57.86644T^2 - 36.50624T^3 - 0.007374/T^2$$

$$C_{pC_4H_{10}O_3} = 90.04$$

$$C_{pCO_2} = 24.99735 + 55.18696T - 33.69137T^2 + 7.948387T^3 - 0.136638/T^2$$

$$C_{pCO} = 25.56759 + 6.096130 + 4.054656T^2 - 2.671301T^3 + 0.131021/T^2$$

$$C_{pN_2} = 28.98641 + 1.852978T - 9.647459T^2 + 16.63537T^3 + 0.000117/T^2$$

comments: the NIST website did not provide data for the Shomate equation for Benzene and Maleic anhydride, so these values were assumed to be constant.

To find the weight of the catalyst, we use:

$$W = \rho_b * \frac{\pi}{4} * D(inner) * L = 3.2 * 10^3 * \frac{\pi}{4} * .01865^2 * 12ft \left(\frac{1m}{3.28ft} \right) = 3.198kg$$

For co-current heat exchanger:

$$\frac{dT_a}{dW} = \frac{Ua(T - T_a)}{\rho_b * m_s * C_{ps}}$$

For counter-current heat exchanger in part 2:

$$\frac{dT_a}{dW} = \frac{Ua(T - T_a)}{m_s * C_{ps}}$$

Where:

$$C_{ps} = 50 + 0.002T + 0.00001T^2 - 4 \times 10^{-9}T^3 \text{ [J/molK]}$$

$$F_s = \frac{m_s}{MM} = \frac{2000 \frac{kg}{hr}}{28 \frac{g}{mol}} \left(\frac{1000g}{1kg} \right) \left(\frac{1hr}{3600s} \right) = 19.8413 \frac{mol}{s}$$

$$m_s = 555.5564 \frac{g}{s}$$

Comments:

After compiling all these equations, I tried to put it into POLYMATH, but it kept giving me an error. I was unable to get the different graphs to plot.

For the second question, I do not have the numbers, but from a theoretical standpoint, a counter-current heat exchanger should be more efficient in exchanging heat with the reactor. This would mean that to achieve a similar conversion to the co-current heat exchanger, less salt would have to be used.

For the third question part a, we must look at the Ergun Equation to determine the effect of changing catalyst diameter on the pressure drop. In the Ergun Equation:

$$\frac{dP}{dW} = - \frac{\frac{G(1-\phi)}{\rho_o * g_c * D_p * \phi^3} \left[150 * \frac{(1-\phi)\mu}{D_p} + 1.75G \right]}{\rho_c A_c} * \frac{P_o}{P} \left(\frac{T}{T_o} \right) \left(\frac{F_T}{F_{To}} \right)$$

we see that the diameter of the catalyst appears twice both time in the denominator of a term. If all other terms are held constant, we can see that an increase in the catalyst diameter would cause a **decrease** in the pressure drop.

For the third question part b, the outlet temperature should increase as the salt inlet temperature increases because the reaction is exothermic. As the outlet temperature increase, the conversion should also increase, so a hotter salt stream should produce a higher conversion. Consideration should be given to the fact that this would also produce a very hot exit stream which could be potentially dangerous.

For the fourth question, there are a few ways to increase conversion. One is to simply use a bigger reactor. If the reactants spend a longer time in the reactor, they have a higher chance of reaction. Obviously, it is not practical to just keep making the reactor bigger, and often the tubes used in reactors come in standard sizes. If a more effective catalyst were to be invented, the conversion would also be increased.

Safety Concerns:

One of the major safety concerns is the possibility of the reaction “running away”. This could happen if the inlet salt temperature were too high cause the outlet temperature to increase rapidly. If the outlet temperature gets too hot, it could create a lot of pressure which could result in an explosion. Another safety concern is the pressure drop in the reactor. If the pressure difference inside of the reactor is too large, the reactor could have large stresses put on it. These stresses could cause welds or other connection to fail. This would mean that the reactor would not be operation and much time and money would have to be spent to repair it.

To address these safety concerns, a thorough thermodynamic evaluation of the process would have to be done to find any potential steady states. This analysis would help determine any ignition points where the reactor temperature could suddenly spike. If any ignition points were found, the inlet salt temperature should be sufficiently low enough that the reactor does not come close to the ignition point. This would mean a lower conversion, but it would come with much more control and safety. Since it is not always possible to control the catalyst diameter, other accommodations would have to be made to make sure the pressure drop does not get too large. One possibility is to increase the diameter of the reactor tube. This would increase the surface area which is in the denominator of the pressure drop equation. In theory this would reduce the pressure drop.